

Week 7 Project Summary

HealthConnect Project

Experience Lab Internship Programme

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1. What I Planned to Test

Per the Week 6 hand-off, the plan was to determine whether the Week 6 candidate model's improvement was statistically real (not single-split noise), whether either candidate showed signs of overfitting, whether the Week 6 cross-track feature actually replicated on unseen data, and whether the Week 6 false-negative blind spot could be reduced through a justified refinement.

2. What I Actually Tested

- Reproducibility of the Logistic Regression v2 candidate (retrained twice, compared results).
- Overfitting, via train-vs-test ROC-AUC gap, for both Week 6 candidates (Logistic Regression v2 and Random Forest).
- Cross-validation stability, re-verifying the Week 6 5-fold result.
- Statistical significance of the Week 5 → Week 6 improvement, using a bootstrap confidence interval on the ROC-AUC difference.
- Error analysis on the refined candidate, to check whether the Week 6 blind spot persisted after feature refinement.
- Out-of-sample replication of the Week 6 cross-track feature (long_lead_followup) on the held-out test set specifically.
- A decision-threshold refinement (0.50 → 0.40), evaluated on both overall metrics and the specific blind-spot segment.

3. Most Important Testing Results

- Reproducibility: Pass — identical results across repeated runs.
- Overfitting: Logistic Regression v2 passes cleanly (train/test gap 0.003); Random Forest shows a larger, borderline gap (0.035) — new evidence favouring Logistic Regression v2 as the primary candidate.
- Statistical significance: the Week 5→Week 6 improvement is real (bootstrap 95% CI [0.0026, 0.0145], excludes zero), even though it is modest in size.

- Cross-track retest: the long_lead_followup finding replicated on held-out data the feature was never trained on (68.4% no-show rate vs. 50% baseline, 1.37x lift) — confirms it is a genuine pattern, not an artefact.
- Threshold refinement: recall on the hardest-to-detect blind-spot segment more than tripled (8.2% → 27.6%) at threshold 0.40, though most cases in that segment are still missed.

4. Issues or Weaknesses Identified

- The short-lead-time/clean-history false-negative blind spot persists even after feature refinement — confirmed to be a threshold/data-availability issue, not something more feature engineering alone can fix.
- Random Forest shows a modest overfitting signal not present in the linear model, a new consideration for the Week 8 final candidate choice.
- The 0.40 threshold recommendation rests on an assumed cost asymmetry (missed no-show costs more than a false alarm) that has not been confirmed with HealthConnect stakeholders.

5. Improvements/Refinements Made

Lowered the operational decision threshold from 0.50 to 0.40 for the Week 6 candidate model. This is a deployment-configuration change, not a retraining — the underlying model is unchanged, but the point at which a predicted probability is treated as “high risk” has shifted to prioritise catching more true no-shows at a measured cost in false alarms.

6. Retesting Results

Full before/after comparison at threshold 0.50 vs. 0.40: recall rose from 65.2% to 81.8%, precision fell from 63.0% to 58.4%, and accuracy was roughly unchanged (63.5% → 61.8%) — confirming accuracy is not the metric that matters for this decision. On the specific blind-spot segment, recall rose from 8.2% to 27.6%. The top-20%-highest-risk business metric (1.44x lift) is unaffected by threshold choice, since it is rank-based, and was reconfirmed unchanged.

7. Track(s) Collaborated With

Data Analytics (retesting a Week 6 cross-track finding)

8. What Was Tested Collaboratively

The Week 6 Data-Analytics-style finding — that Follow-up appointments are the most lead-time-sensitive type — was retested specifically on the held-out test set (data the finding was not derived from), rather than simply re-communicated. This satisfies the Week 7 requirement

that collaboration produces a Test → Finding → Action → Retest → Validated Improvement chain, not just discussion.

9. What Changed as a Result

The long_lead_followup feature now carries out-of-sample validation evidence, not just Week 6 training-set feature importance — it can be reported to Data Analytics as confirmed, and to ML Engineering as a stable, tested pipeline input rather than a candidate for removal if simplification is needed.

10. Key Findings or Validation Outcomes

- The Week 6 model-family choice (Random Forest vs. Logistic Regression) doesn't matter much — but the Week 6 feature refinements demonstrably do (statistically confirmed this week).
- The model's real limitation isn't discrimination power in aggregate (ROC-AUC ~0.68–0.69) — it's a specific, now well-characterised blind spot that threshold tuning only partially addresses.
- The model is validated as suitable for risk-based prioritisation (e.g. top-20% outreach targeting), not for high-stakes individual decisions.

11. Major Challenges

- Distinguishing a real fix from a cosmetic one — confirming that the blind spot is a threshold/data-availability limitation (not a feature-engineering gap) required deliberately testing a segment-specific metric, not just the overall recall figure.
- Setting a defensible threshold without a stated business cost ratio from HealthConnect — addressed by making the underlying assumption explicit and flagging it for stakeholder confirmation rather than silently picking a number.

12. Important Decisions

- Chose to recommend Logistic Regression v2 over Random Forest as the primary Week 8 candidate, now further supported by the overfitting-gap evidence found this week, on top of Week 6's cross-validation tie.
- Chose a threshold change (0.40) over further feature engineering to address the blind spot, since error analysis showed the issue was about decision sensitivity, not missing signal.
- Chose to report the blind-spot fix as partial rather than resolved — 27.6% recall on that segment is a real improvement over 8.2%, but still means most of those cases are missed.

13. Remaining Limitations

- Most short-lead/clean-history no-shows remain undetected even after threshold tuning.
- Fairness/bias testing across gender and age subgroups remains outstanding — now the top priority before Week 8.

- The 0.40 threshold has not been validated against an actual HealthConnect cost model.

14. Remaining Issues or Dependencies

- Stakeholder sign-off on the threshold trade-off (via Project Management).
- Formal hand-off of the final feature list, scaler, model, and threshold to Machine Learning Engineering for pipeline integration.
- Data Dictionary formal cross-check, outstanding since Week 4 (low risk, not yet closed).

15. Contribution to the Overall HealthConnect Project

This week converts the Week 6 candidate model from “validated but untested” to “tested, statistically checked, and partially refined against a known weakness” — with an explicit, honest account of what still doesn't work. The cross-track retest gives the Data Analytics track independently-confirmed evidence for their own Week 7 output, and the finalised, tested model configuration (features, scaler, threshold) is now ready to hand off to Machine Learning Engineering.

16. What Must Be Completed Before Week 8

- Fairness/bias testing across demographic subgroups.
- Stakeholder confirmation of the 0.40 threshold's underlying cost assumption.
- Formal hand-off package for Machine Learning Engineering.
- Honest documentation of the accepted blind spot for the Week 8 final presentation.